ATOM-SG Pilot - Week 1 Baseline Test

P6 Mathematics | 40 Questions | 50 Minutes | No Calculator

Name: _____ **Class:** _____ **Date:** _____

Instructions: Show all working. Professional-grade diagrams included with Geometry questions.

Q 1. [E] [WP] Before-After

A shop had 240 pens. On Monday, they sold 2/5 of the pens. On Tuesday, they sold 1/3 of the remaining pens. How many pens were left?

Working:

Answer: _____

Q 2. [M] [WP] Before-After

Sarah spent 3/8 of her money on a book and 2/5 of the remainder on a pen. If she had $30 left, how much did she have at first?

Working:

Answer: _____

Q 3. [E] [WP] Part-Whole

Ali has $120. Ben has 3/4 as much as Ali. How much money do they have altogether?

Working:

Answer: _____

Q 4. [M] [WP] Part-Whole

The ratio of boys to girls in a class is 5:3. If there are 40 students altogether, how many boys are there?

Working:

Answer: _____

Q 5. [M] [WP] Cross-Thread

A baker made 360 cupcakes. He sold 1/4 of them in the morning and 2/3 of the remainder in the afternoon. How many cupcakes did he have left?

Working:

Answer: _____

Q 6. [H] [WP] Cross-Thread

Alice and Betty had $280 altogether. After Alice spent 1/3 of her money and Betty spent $40, they had the same amount left. How much did Alice have at first?

Working:

Answer: _____

Baseline Test - Continued

Q 7. [M] [WP] Before-After

A tank was full of water. After 3/7 of the water was poured out, then 2/5 of the remaining water was used, 48 litres were left. How much water was in the tank at first?

Working:

Answer: _____

Q 8. [M] [WP] Part-Whole

John has 3/5 as many stamps as Mary. Mary has 5/6 as many stamps as Tom. If Tom has 72 stamps, how many stamps does John have?

Working:

Answer: _____

Q 9. [E] [WP] Before-After

A bus had 60 passengers. At the first stop, 15 passengers got off. At the second stop, 8 passengers got off. How many passengers were left?

Working:

Answer: _____

Q10. [H] [WP] Cross-Thread

The ratio of red balls to blue balls in a bag was 3:5. After removing 12 red balls, the ratio became 1:3. How many red balls were there at first?

Working:

Answer: _____

Q11. [H] [WP] Cross-Thread

The ratio of Alan's age to Ben's age is 4:7. In 6 years, the ratio will be 2:3. How old is Alan now?

Working:

Answer: _____

Q12. [M] [WP] Before-After

A library had 450 books. They received 120 new books on Monday. On Tuesday, they donated 1/5 of their total books to a school. How many books did they have left?

Working:

Answer: _____

Baseline Test - Continued

Q13. [E] [WP] Part-Whole

There are 80 students in a hall. 3/8 of them are girls. How many boys are there?

Working:

Answer: _____

Q14. [M] [WP] Cross-Thread

A fruit seller had 240 apples. He sold 3/4 of them to Shop A and 1/2 of the remainder to Shop B. How many apples did he have left?

Working:

Answer: _____

Q15. [M] [WP] Part-Whole

Jenny and Kelly have 84 stickers altogether. Jenny has 16 more stickers than Kelly. How many stickers does Kelly have?

Working:

Answer: _____

Q16. [H] [WP] Before-After

A shop increased the price of a bag by 20%. During a sale, they offered a 15% discount on the new price. If the final price was $204, what was the original price?

Working:

Answer: _____

Q17. [H] [WP] Part-Whole

The ratio of apples to oranges to pears is 2:3:5. If there are 120 oranges and pears altogether, how many apples are there?

Working:

Answer: _____

Q18. [H] [WP] Cross-Thread

A tank was 5/6 full of water. After 15 litres were poured out, it was 2/3 full. What is the capacity of the tank?

Working:

Answer: _____

Q19. [M] [WP] Before-After

Mrs. Tan had some money in her purse. She spent $25 on groceries and then deposited 1/4 of the remainder into her bank account. If she had $45 left, how much did she have at first?

Working:

Answer: _____

Q20. [M] [WP] Part-Whole

The ratio of boys to girls in a club is 5:3. There are 24 more boys than girls. How many children are in the club?

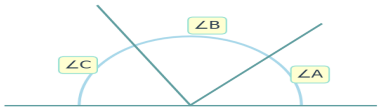
Working:

Answer: _____

Q21. [E] [G] Geometry-Angle

Measure the angles A, B, and C in the diagram using a protractor.

G001: Measure angles with protractor



Measure angles ∠A, ∠B, and ∠C in the diagram

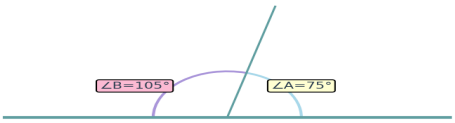
Working:

Answer: _____

Q22. [E] [G] Geometry-Angle

Two angles on a straight line are given. If one angle is 75°, find the other angle.

G002: Angles on straight line



Two angles on a straight line

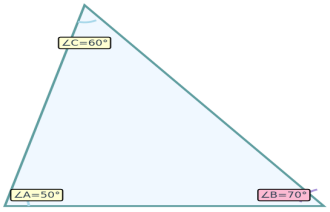
Working:

Answer: _____

Q23. [M] [G] Geometry-Triangle

In triangle ABC, angle A = 50° and angle B = 70°. Find angle C.

G005: Triangle angle sum



Triangle ABC with two angles given

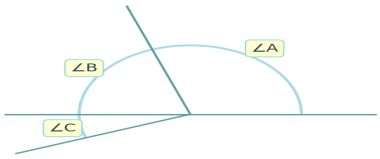
Working:

Answer: _____

Q24. [H] [G] Geometry-Angle

Angle A and Angle B are on a straight line. Angle B and Angle C are complementary. If Angle A = 110°, find Angle C.

G007: Combined angle properties



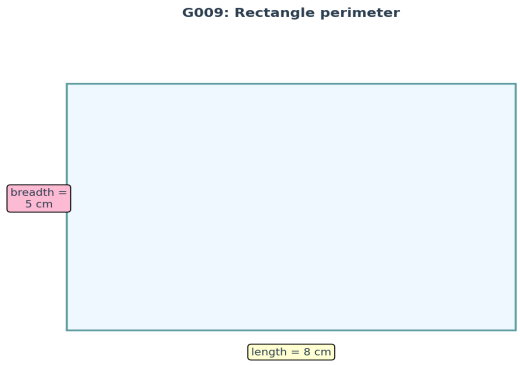
Two-step angle reasoning

Working:

Answer: _____

Q25. [E] [G] Geometry-Perimeter

Find the perimeter of the rectangle shown in the diagram.

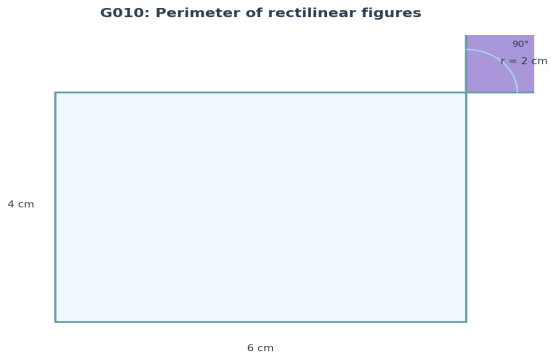


Working:

Answer: _____

Q26. [M] [G] Geometry-Composite

A rectilinear figure consists of two rectangles joined together. Find its perimeter.



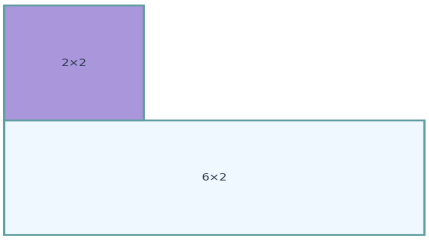
Working:

Answer: _____

Q27. [M] [G] Geometry-Area

Find the area of an L-shape figure formed by two rectangles.

G011: Area of L-shape



Working:

Answer: _____

Q28. [M] [G] Geometry-Conversion

Convert 2.5 square metres to square centimetres.

G014: Area unit conversion

2.5 m^2 = 25,000 cm^2

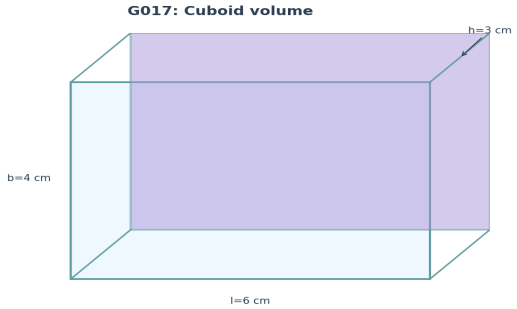
Area unit conversion

Working:

Answer: _____

Q29. [E] [G] Geometry-Volume

Find the volume of the cuboid shown in the diagram.

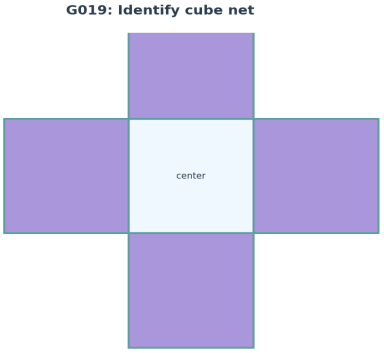


Working:

Answer: _____

Q30. [M] [G] Geometry-3D

Which of the following nets can be folded to form a cube?

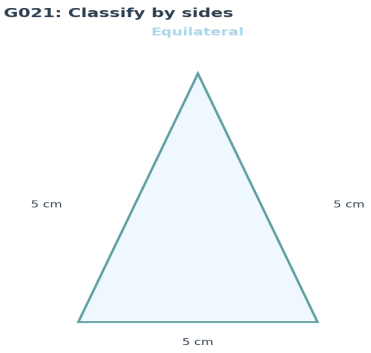


Working:

Answer: _____

Q31. [E] [G] Geometry-Classify

Classify the triangle with side lengths 5 cm, 5 cm, 5 cm.

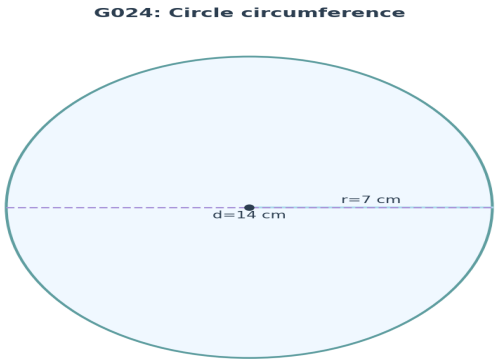


Working:

Answer: _____

Q32. [M] [G] Geometry-Circle

A circle has radius 7 cm as shown. Find its circumference (take $\pi = 22/7$).

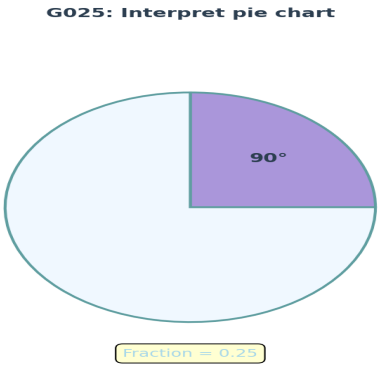


Working:

Answer: _____

Q33. [E] [DI] DI-Pie Chart

The pie chart shows fruit distribution. What fraction represents the 72° sector?

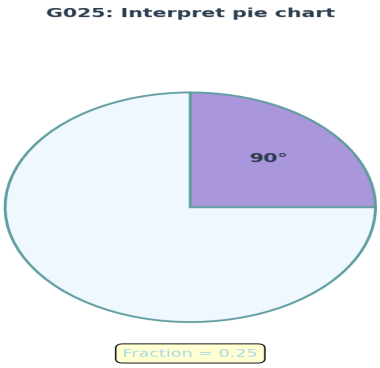


Working:

Answer: _____

Q34. [E] [DI] DI-Pie Chart

If 72° represents 60 apples, how many fruits are there in total?

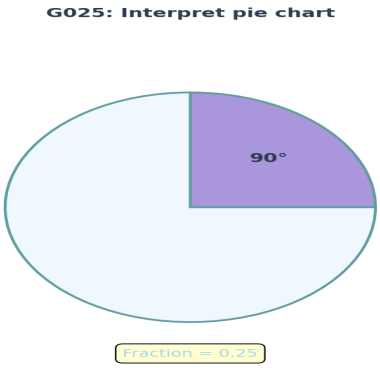


Working:

Answer: _____

Q35. [M] [DI] DI-Pie Chart

Calculate the percentage for each fruit type in the pie chart.



Working:

Answer: _____

Q36. [M] [DI] DI-Chart

The chart shows survey results. Which category has the highest percentage?

Working:

Answer: _____

Q37. [M] [DI] DI-Chart

Calculate the difference between the largest and smallest sectors.

Working:

Answer: _____

Q38. [H] [DI] DI-Chart

If the total number of respondents was 500, how many chose each category?

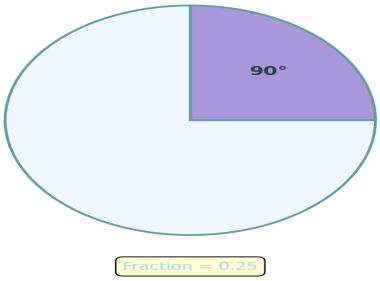
Working:

Answer: _____

Q39. [M] [DI] DI-Pie Chart

Convert each sector angle to a fraction of the whole circle.

G025: Interpret pie chart



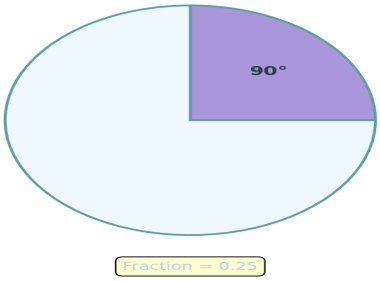
Working:

Answer: _____

Q40. [H] [DI] DI-Pie Chart

If oranges represent 90° and there are 75 oranges, find the total number of fruits.

G025: Interpret pie chart



Working:

Answer: _____

Answer Key - For Markers Only

Q 1: 96 pens
Q 2: $80
Q 3: $210
Q 4: 25 boys
Q 5: 90 cupcakes
Q 6: $144
Q 7: 140 litres
Q 8: 36 stamps
Q 9: 37 passengers
Q10: 27 red balls
Q11: 24 years old
Q12: 456 books
Q13: 50 boys
Q14: 30 apples
Q15: 34 stickers
Q16: $200
Q17: 30 apples
Q18: 90 litres
Q19: $85
Q20: 96 children
Q21: See rubric
Q22: 105°
Q23: 60°
Q24: 20°
Q25: 26 cm
Q26: See rubric
Q27: See rubric
Q28: 25,000 cm^2
Q29: 72 cm^3
Q30: See rubric
Q31: Equilateral
Q32: 44 cm
Q33: 1/5
Q34: 300 fruits
Q35: See working
Q36: See chart
Q37: See working
Q38: See working
Q39: See working
Q40: 300 fruits